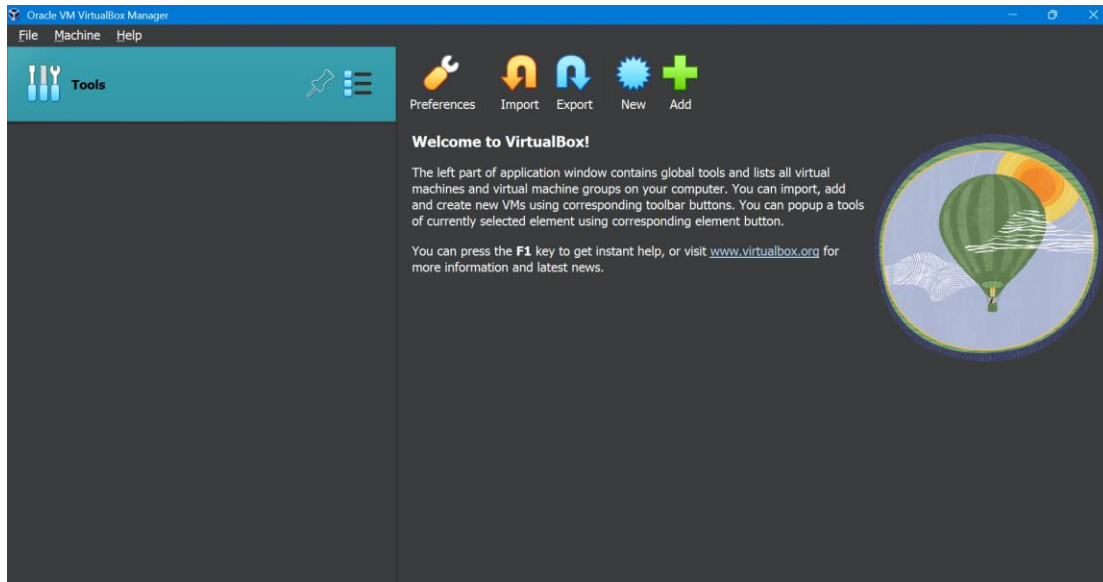


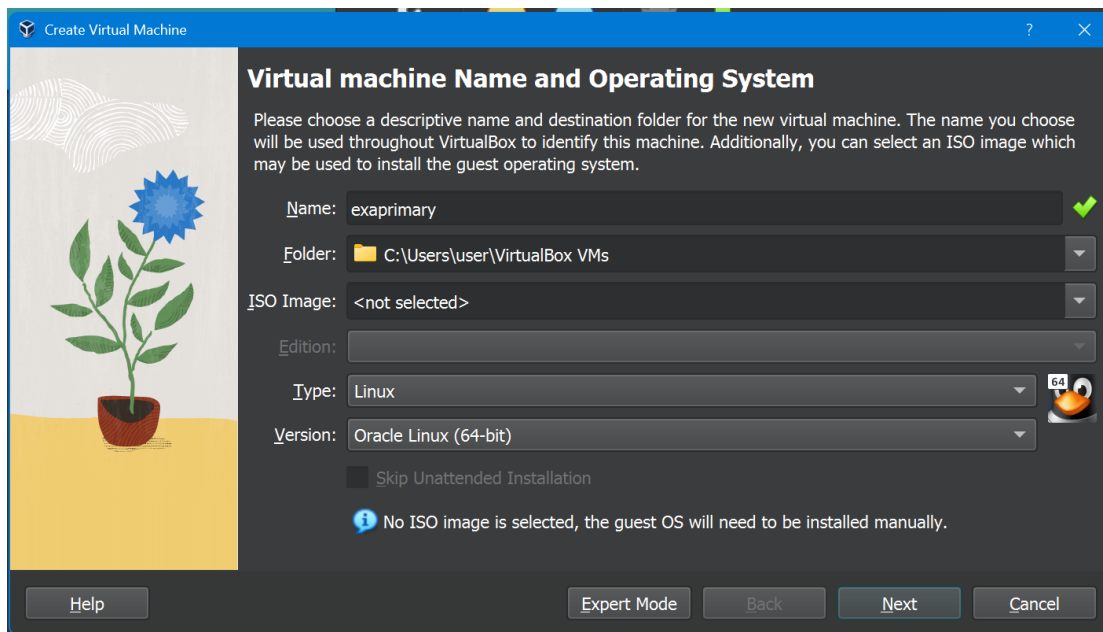
INSTALLATION ORACLE 19C (LINUX)

Oracle VM Virtual Box

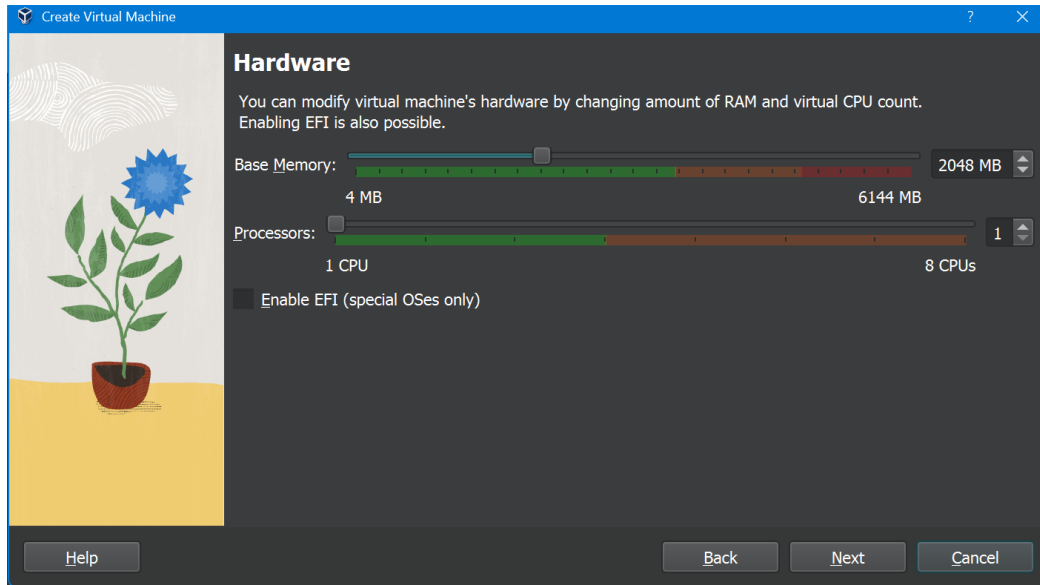
1. Klik New, untuk membuat Servernya



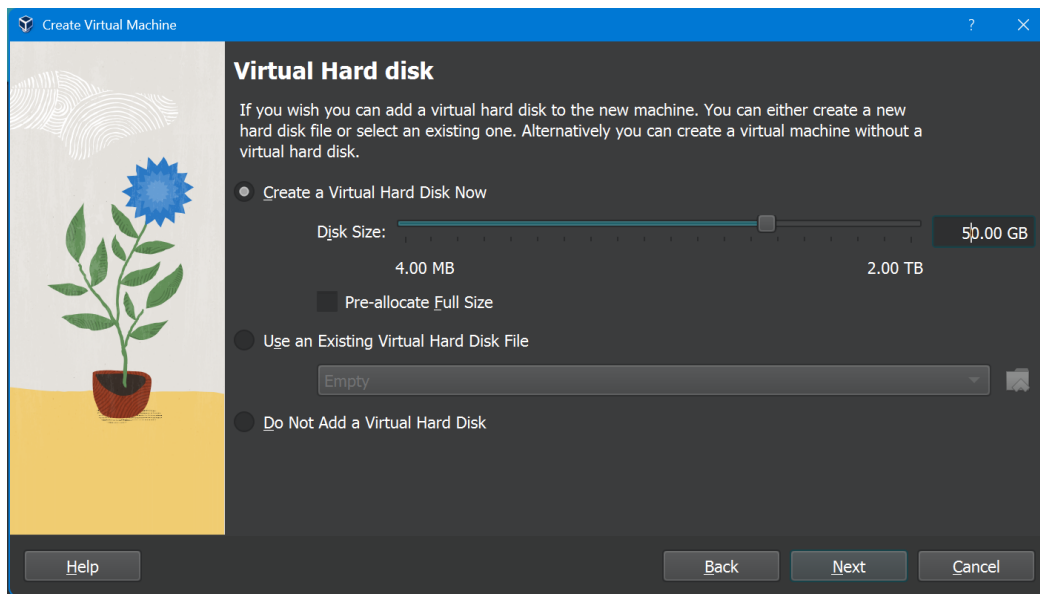
2. Berikan nama Server, Type = Linux, Version = Oracle Linux (64-bit)



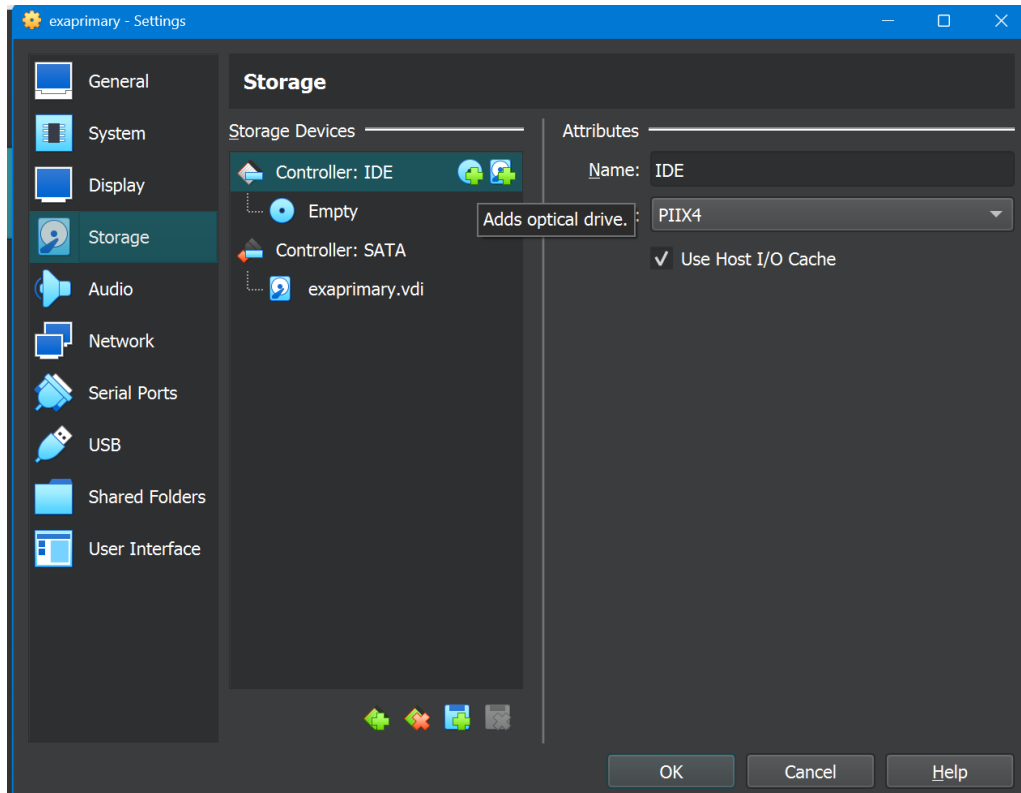
3. Tentukan ukuran RAM yang ini digunakan (opsional)



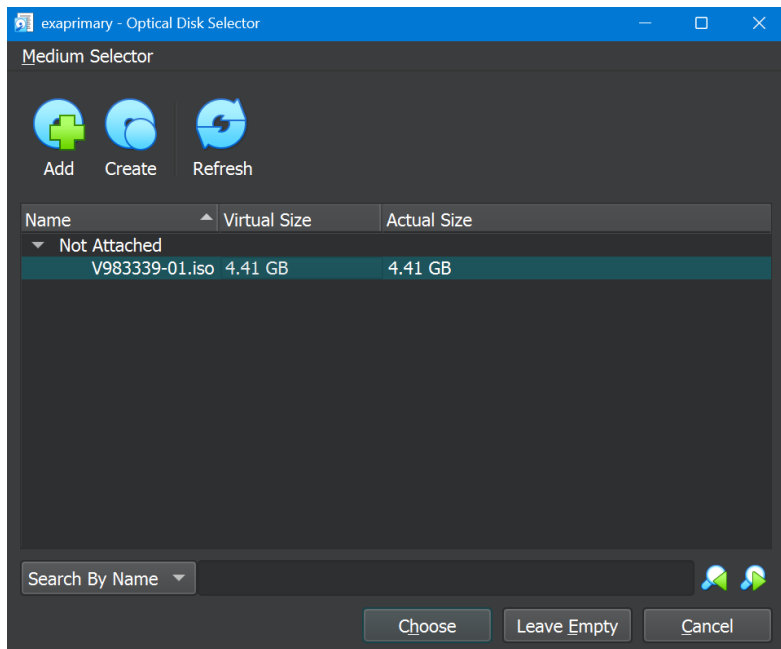
4. Tentukan Memori Hard Disk yang ingin digunakan (opsional)



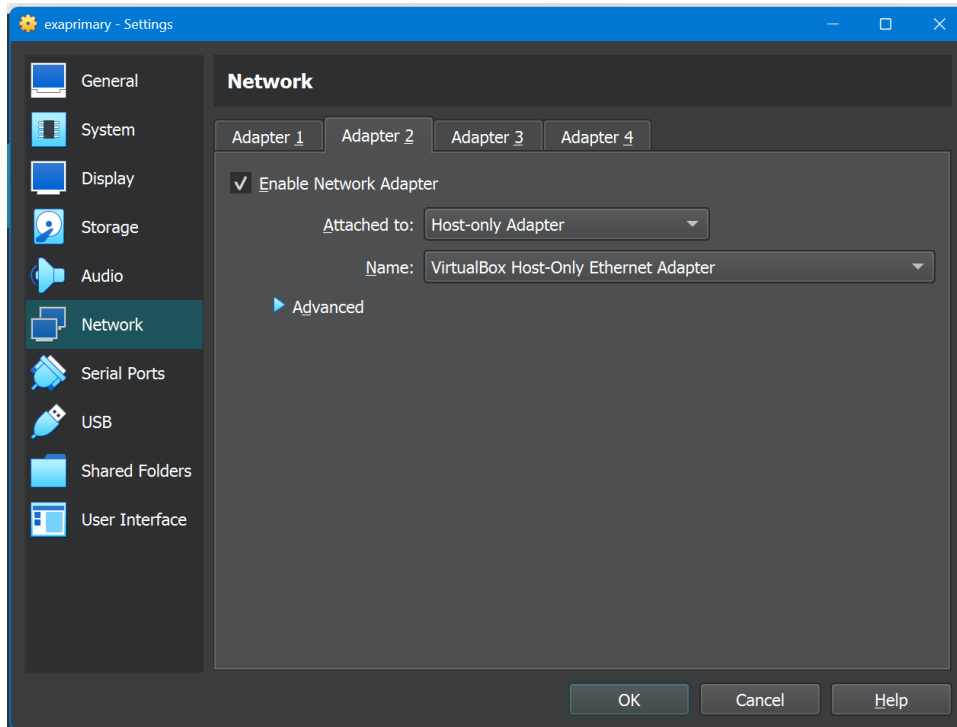
5. Setelah selesai membuat servernya, masuk ke settingan servernya untuk mengatur beberapa, disini diharuskan mengupload file .iso



6. Pilih file .iso nya

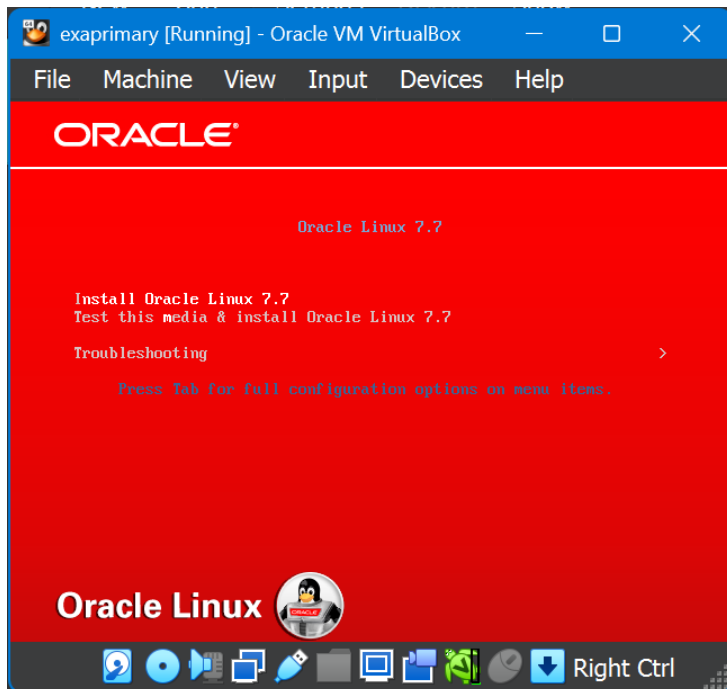


7. setting pada bagian network, ke adapter 2 lalu jadikan host-only adapter

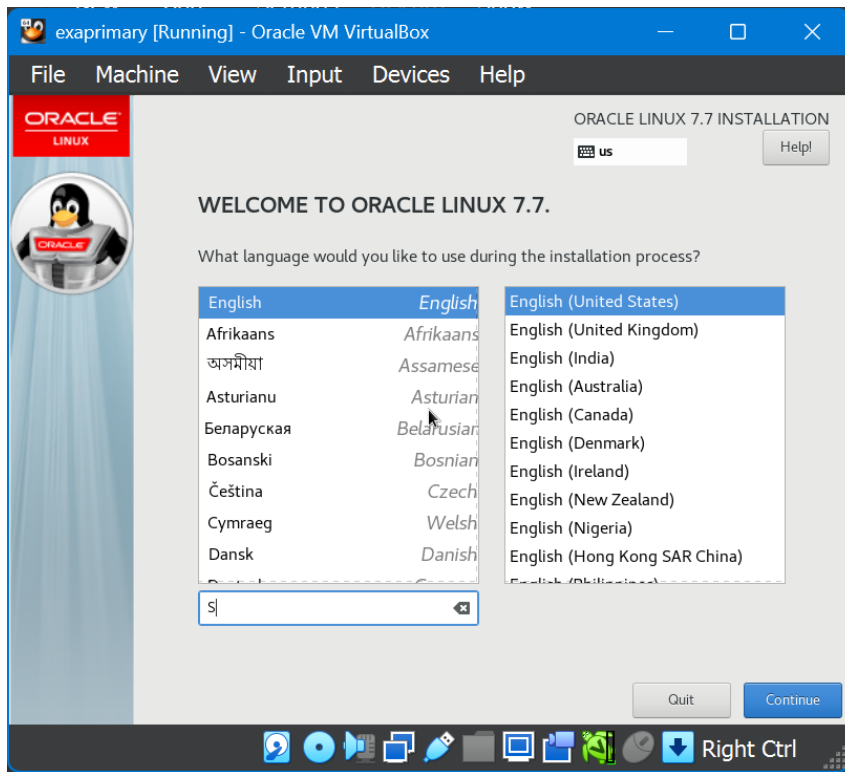


8. Lalu Setelah selesai semua klik 'OK', dan mulai servernya klik 'Start'

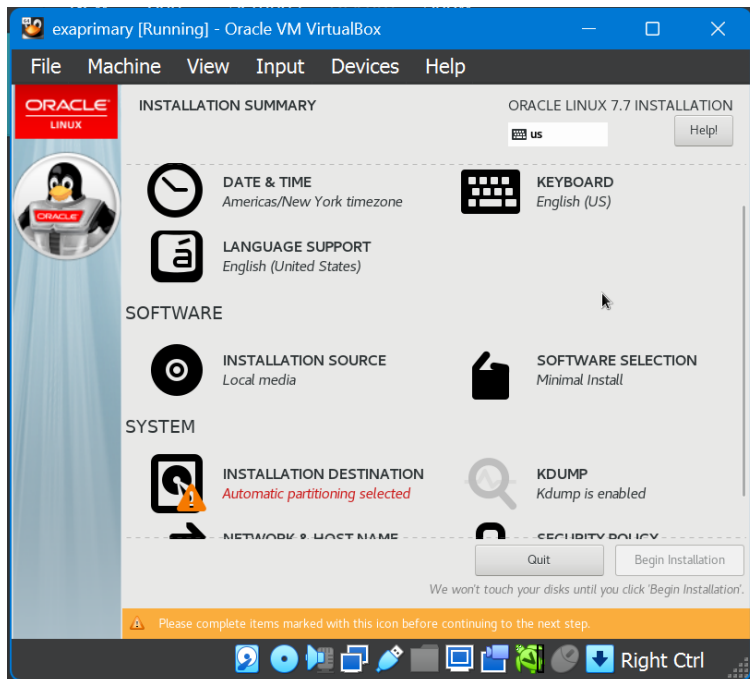
9. Pilih Install Oracle Linux 7.7 lalu enter



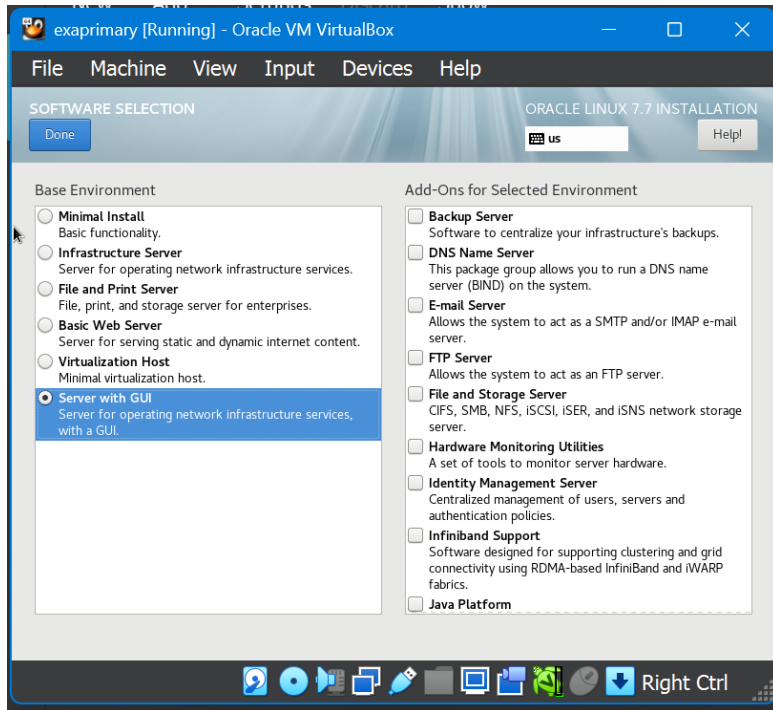
10. Masukan bahasa



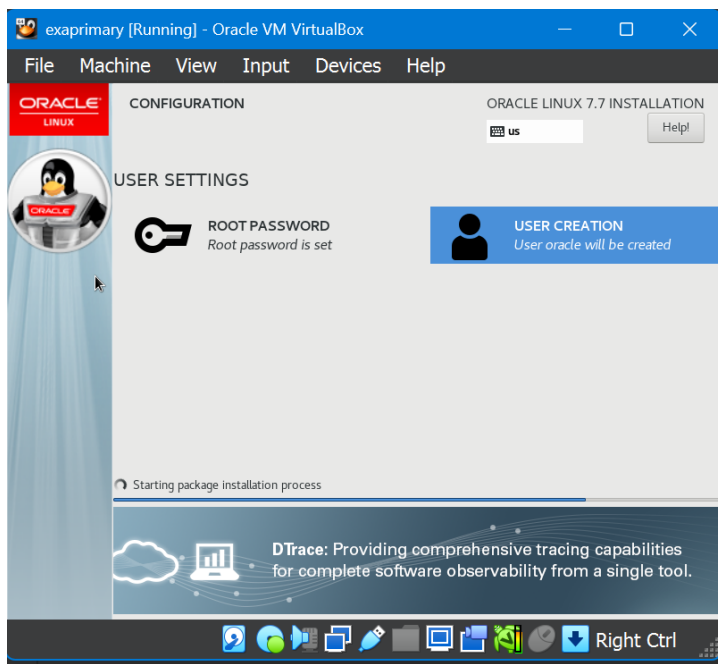
Isi Date Time sesuai dengan waktu dan tempat kalian



Atur User Creation menjadi 'Server with GUI'



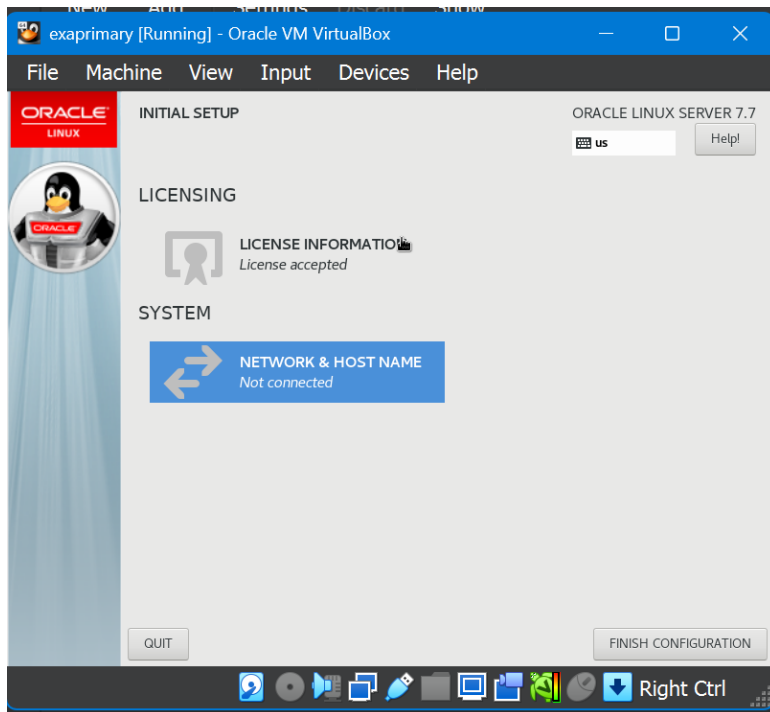
Membuat password dan nama usercreationnya (wajib hafal passwordnya),
Sisanya sesuai default saja.



Setelah selesai Membuat password dan user creationnya, tunggu instalasi selesai.

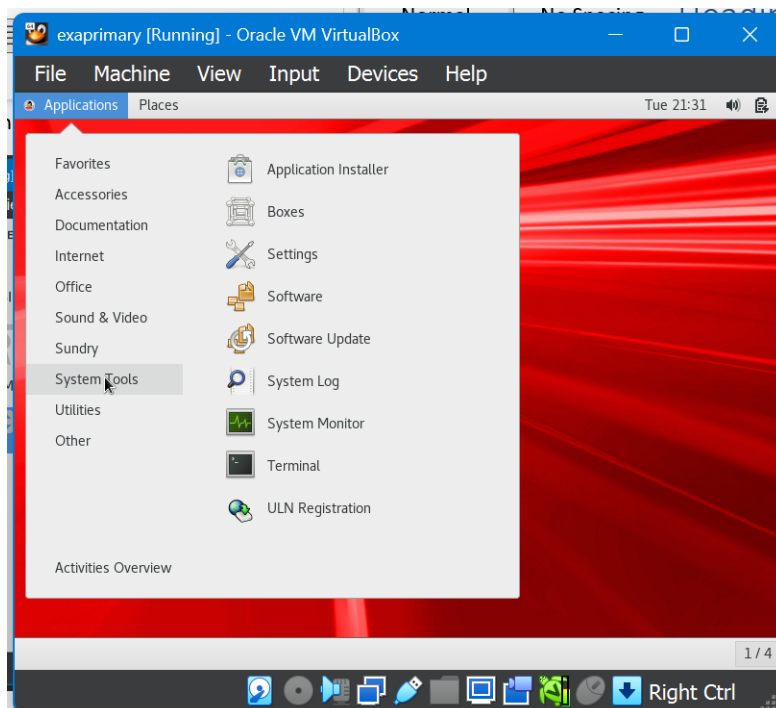
11. Selesai, klik 'reboot'

12. Accept License Information, Lalu Finish Configuration

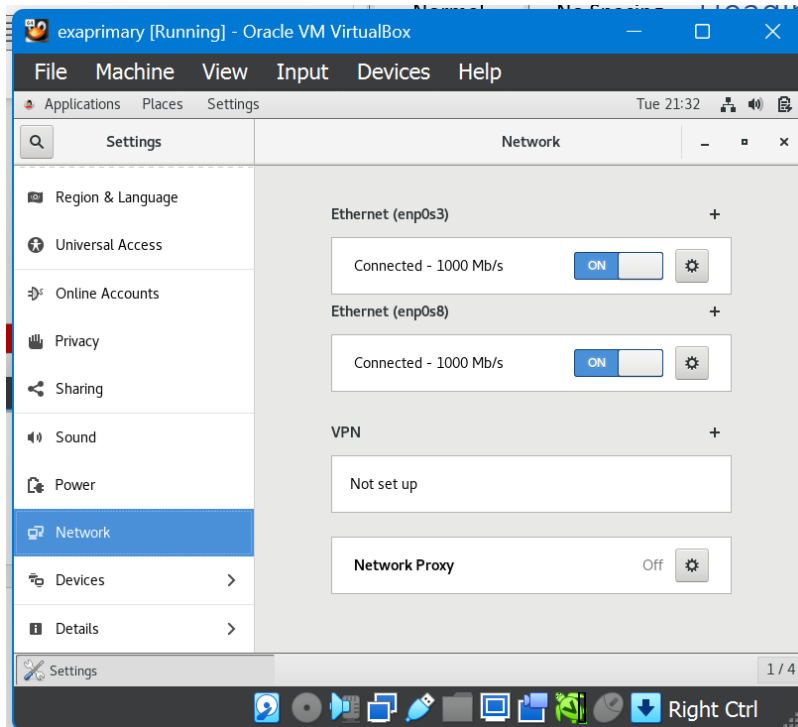


13. Login Servernya

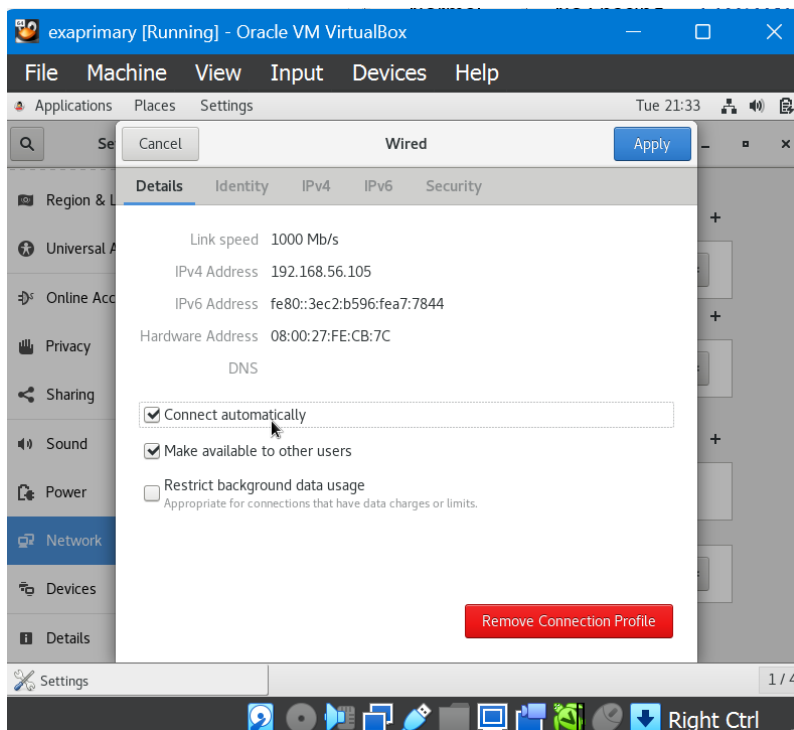
14. Setelah Login, setting networknya

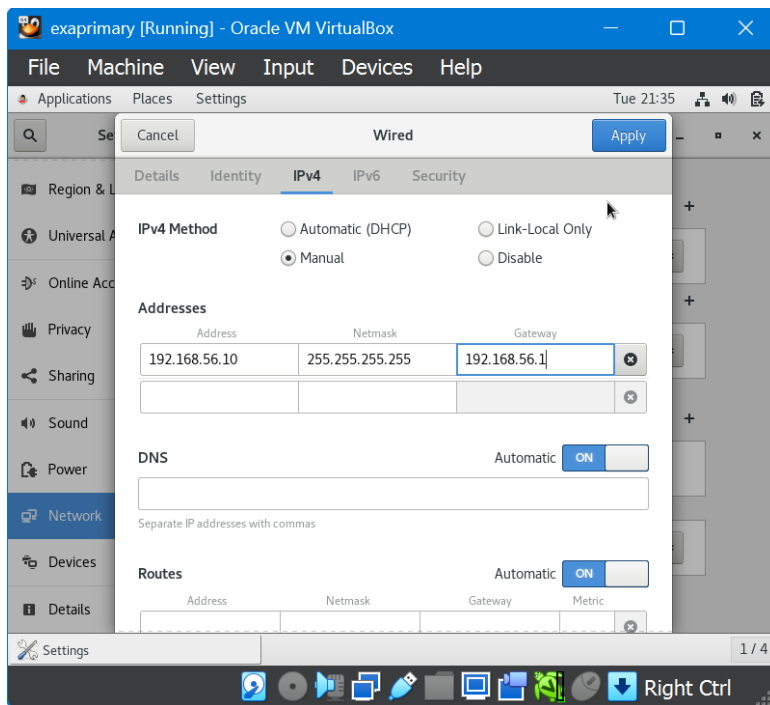


Aktifkan enp0s3 dan enp0s8



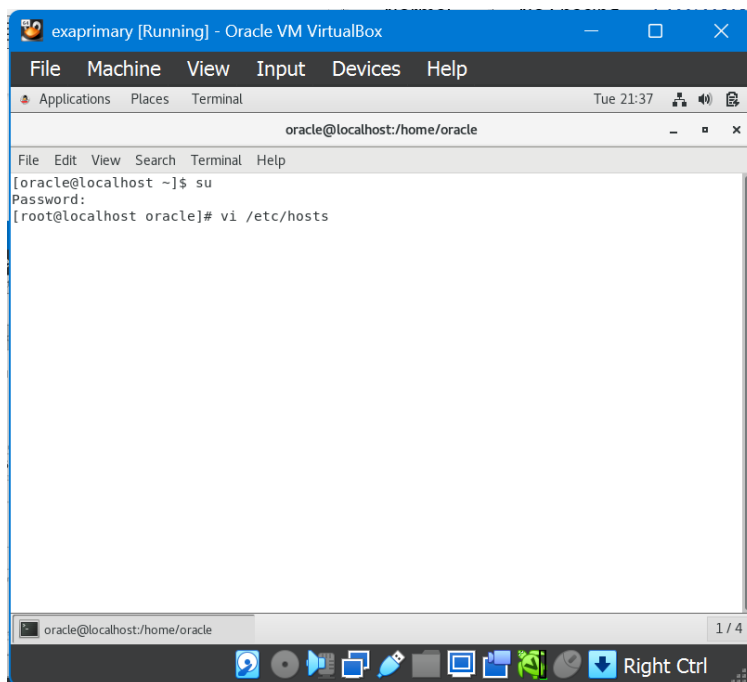
Pada enp0s8, aktifkan connect automatically dan setting Alamat servernya



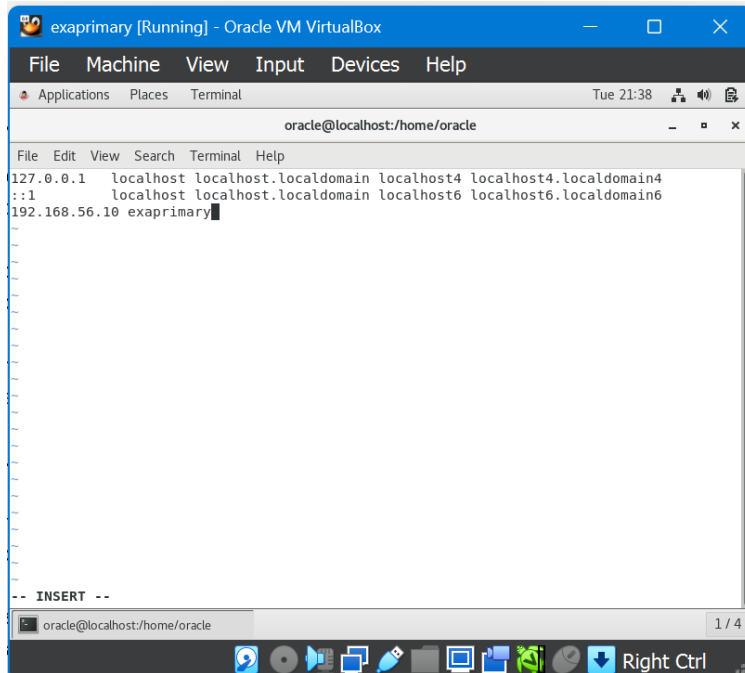


Setelah dari setting, masuk ke terminal dan masuk ke user root, lalu masukan Alamat server kita yang tadi sudah kita setting agar terhubung servernya

Vi /etc/hosts



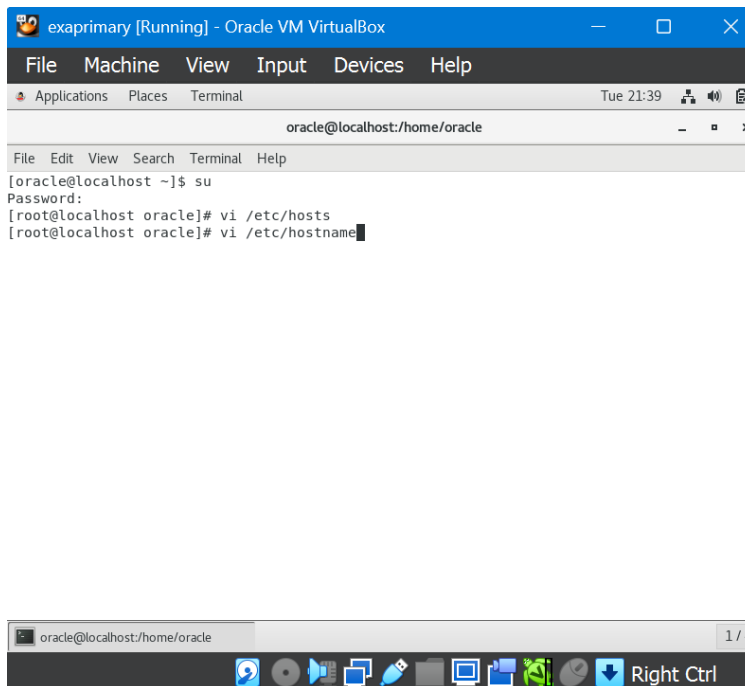
Masukkan Alamat dan nama servernya



```
exaprimary [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Applications Places Terminal Tue 21:38
oracle@localhost:/home/oracle
File Edit View Search Terminal Help
127.0.0.1 localhost localhost.localdomain localhost4 localhost4.localdomain4
::1 localhost localhost.localdomain localhost6 localhost6.localdomain6
192.168.56.10 exaprimary
-- INSERT --
oracle@localhost:/home/oracle 1 / 4
```

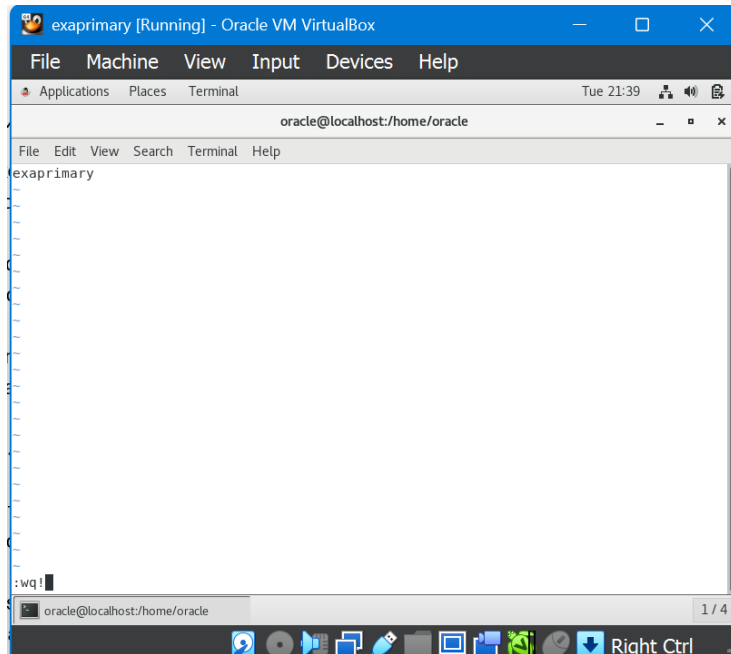
Lalu hubungkan juga di

vi /etc/hostname

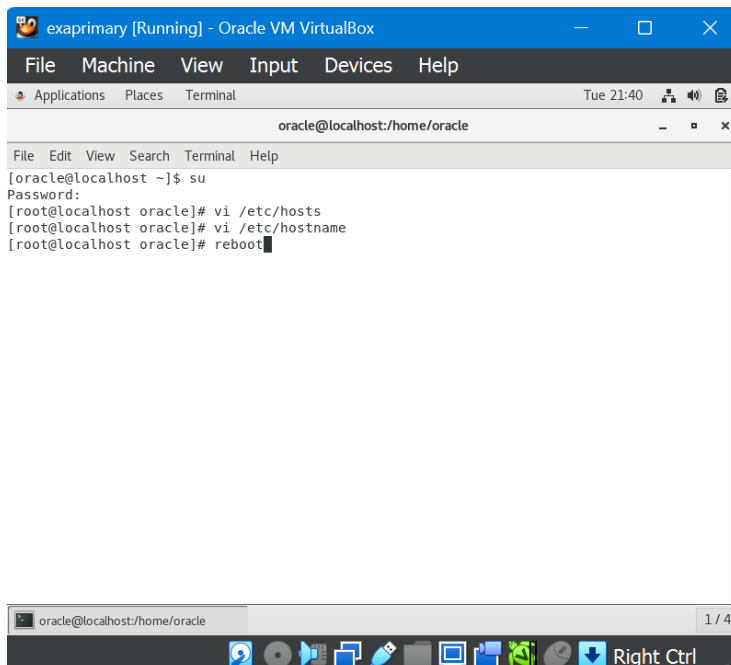


```
exaprimary [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Applications Places Terminal Tue 21:39
oracle@localhost:/home/oracle
File Edit View Search Terminal Help
[oracle@localhost ~]$ su
Password:
[root@localhost oracle]# vi /etc/hosts
[root@localhost oracle]# vi /etc/hostname
oracle@localhost:/home/oracle 1 / 4
```

Isi dengan nama server yang kamu buat

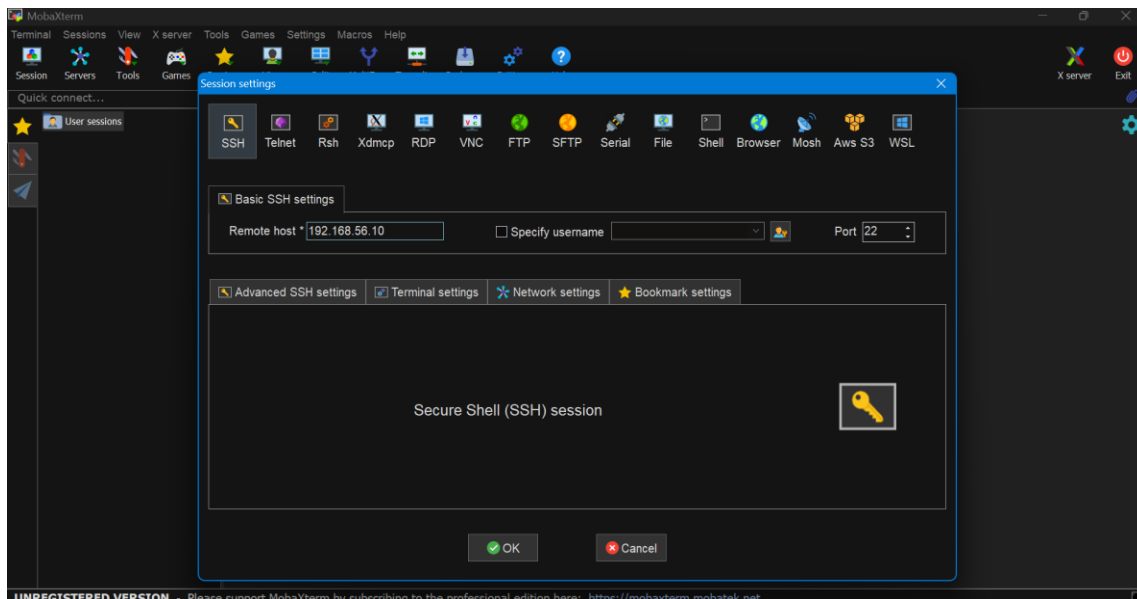


Setelah selesai semua, reboot



15. Lalu kita mulai masuk keserver yang kita buka dengan menggunakan MobaXtream

Klik new session dan pilih SSH, masukkan Alamat servernya lalu 'OK'



Lalu masuk ke servernya dengan memasukan nama user dan password yang sudah dibuat diawal tadi

16. masuk ke server root

```
su - root / su
```

17. mengupload dba.tar dan file zip linux

18. AUTOMATIC SETUP (Install set up an oracle 19c)

```
- yum install -y oracle-database-preinstall-19c
```

19. Setelah proses preinstall selesai lalu lakukan update pada server tersebut.

- yum update -y

Catatan: gunakan automatic setup jika hanya digunakan untuk lab

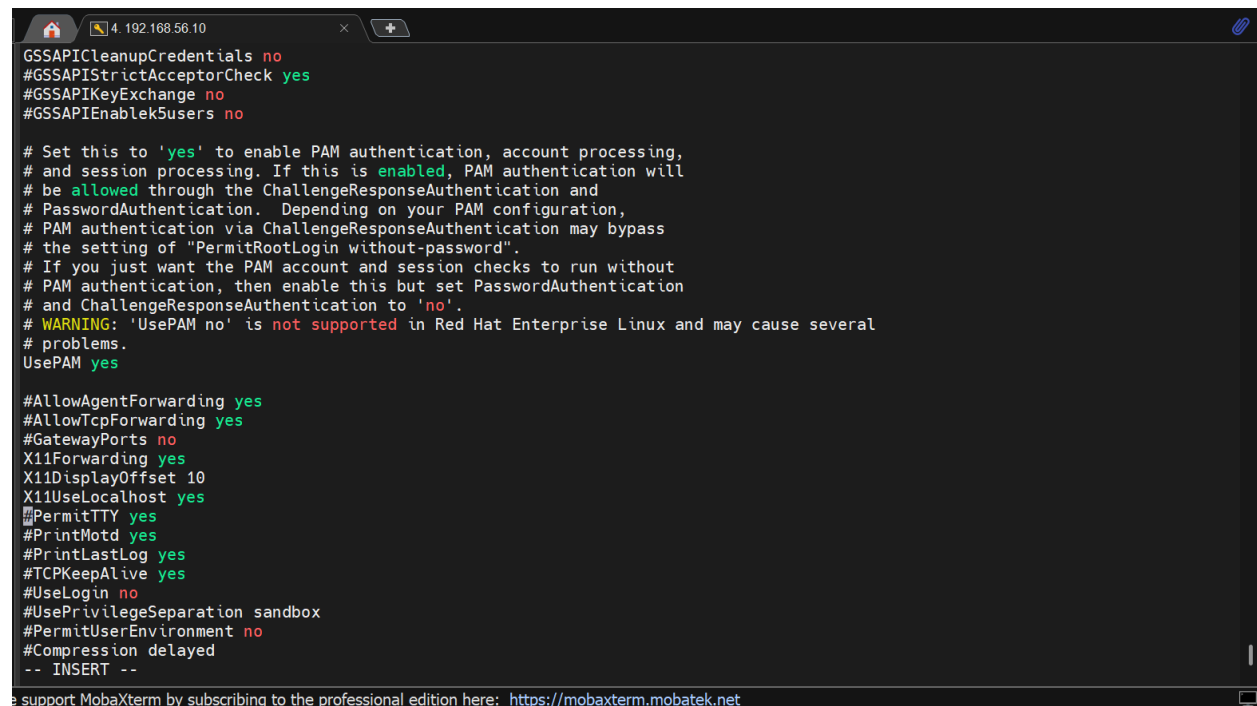
20. AKTIFKAN X11

Setelah selesai memasukkan script diatas selanjut nya kita install x11

- yum install xorg-x11-xauth xorg-x11-fonts* -y

masuk ke filenya dengan menggunakan command

- vi /etc/ssh/sshd_config



```
GSSAPICleanupCredentials no
#GSSAPIStrictAcceptorCheck yes
#GSSAPIKeyExchange no
#GSSAPIEnableK5Users no

# Set this to 'yes' to enable PAM authentication, account processing,
# and session processing. If this is enabled, PAM authentication will
# be allowed through the ChallengeResponseAuthentication and
# PasswordAuthentication. Depending on your PAM configuration,
# PAM authentication via ChallengeResponseAuthentication may bypass
# the setting of "PermitRootLogin without-password".
# If you just want the PAM account and session checks to run without
# PAM authentication, then enable this but set PasswordAuthentication
# and ChallengeResponseAuthentication to 'no'.
# WARNING: 'UsePAM no' is not supported in Red Hat Enterprise Linux and may cause several
# problems.
UsePAM yes

#AllowAgentForwarding yes
#AllowTcpForwarding yes
#GatewayPorts no
X11Forwarding yes
X11DisplayOffset 10
X11UseLocalhost yes
#PermitTTY yes
#PrintMotd yes
#PrintLastLog yes
#TCPKeepAlive yes
#UseLogin no
#UsePrivilegeSeparation sandbox
#PermitUserEnvironment no
#Compression delayed
-- INSERT --
```

support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.net>

Lalu hapus # pada bagian x11

21. Setelah itu lakukan

- systemctl restart sshd

```
[root@exapprimary oracle]# vi /etc/ssh/sshd_config
[root@exapprimary oracle]# systemctl restart sshd
[root@exapprimary oracle]#
```

22. Install command dibawah ini :

yum install -y bc

yum install -y binutils

yum install -y compat-libcap1

yum install -y compat-libstdc++-33

#yum install -y dtrace-modules

#yum install -y dtrace-modules-headers

#yum install -y dtrace-modules-provider-headers

yum install -y dtrace-utils

yum install -y elfutils-libelf

yum install -y elfutils-libelf-devel

yum install -y fontconfig-devel

yum install -y glibc

yum install -y glibc-devel

yum install -y ksh

yum install -y libaio

yum install -y libaio-devel

yum install -y libdtrace-ctf-devel

yum install -y libXrender

yum install -y libXrender-devel

yum install -y libX11

yum install -y libXau

```
yum install -y libXi
yum install -y libXtst
yum install -y libgcc
yum install -y librdmacm-devel
yum install -y libstdc++
yum install -y libstdc++-devel
yum install -y libxcb
yum install -y make
yum install -y net-tools # Clusterware
yum install -y nfs-utils # ACFS
yum install -y python # ACFS
yum install -y python-configshell # ACFS
yum install -y python-rtlib # ACFS
yum install -y python-six # ACFS
yum install -y targetcli # ACFS
yum install -y smartmontools
yum install -y sysstat
yum install -y gcc
yum install -y unixODBC
```

23. lalu buat grup and users

```
groupadd -g 54321 oinstall
groupadd -g 54322 dba
groupadd -g 54323 oper
#groupadd -g 54324 backupdba
#groupadd -g 54325 dgdba
```


26. Setelah selesai jika firewall masih enabled, kalian harus disable dengan command dibawah ini

- systemctl stop firewalld
- systemctl disable firewalld

27. create directory in the wich the oracle software will be installed

```
mkdir -p /u01/app/oracle/product/19.0.0/dbhome_1
```

```
mkdir -p /u02/oradata
```

```
chown -R oracle:oinstall /u01 /u02
```

```
chmod -R 775 /u01 /u02
```

28. Lalu masuk ke server oracle

- su – oracle

29. Ekstrak file dba.tar di home directory dari oracle user di target database host

- tar -xvf dba.tar

30. masuk file

```
cd dba
```

31. menyalin (copy) sebuah file atau direktori bernama act ke dalam folder tujuan /home/oracle.

```
cp act /home/oracle
```

32. membuka dan mengedit file konfigurasi lingkungan (environment) database menggunakan text editor vi.

- vi .db_profile

[illegible]

```
[oracle@exapprimary dba]$ vi .db_profile
[oracle@exapprimary dba]$ vi .db_profile
[oracle@exapprimary dba]$ cd
[oracle@exapprimary ~]$ . ./act
[oracle@exapprimary ~]$ sett
List Oracle SID & Oracle Home :
[1] orcl /u01/app/oracle/product/19.0.0/dbhome_1
Choose SID [1]: 1
ORACLE SID is set to orcl
```

```
chmod -R 775 /u01
```

35. Cari Lokasi Pasti File .zip

Ketik perintah ini (sebagai root atau oracle) untuk menemukan "koordinat" tepat file tersebut:

su - oracle

```
find / -name "LINUX.X64_193000_db_home.zip" 2>/dev/null
```

36. Masuk ke Folder Tujuan

Langkah ini wajib agar hasil ekstrak tidak berantakan lagi:

```
- cd /u01/app/oracle/product/19.0.0/dbhome_1/
```

37. Jalankan Unzip dengan Path Lengkap

Gunakan perintah ini untuk menarik file dari folder home kamu ke folder tujuan:

```
- unzip -oq /home/oracle/LINUX.X64_193000_db_home.zip
```

38. Instal Paket X11 (Sebagai root)

Pindah ke root

su - root

Instal xclock dan library pendukungnya

```
yum install -y xorg-x11-apps
```

39. Pastikan X11 Forwarding Aktif di MobaXterm

1. Klik kanan pada nama Session kamu di panel kiri MobaXterm -> pilih Edit Session.
2. Pergi ke tab Advanced SSH settings.
3. Pastikan kotak X11-Forwarding sudah dicentang (checked).
4. Klik OK, lalu Restart Session (tutup terminal dan buka lagi).

40. Set Manual Tanpa xhost (Sebagai User oracle)

Jangan gunakan root, langsung saja ke user oracle. Jalankan perintah ini:

```
su - oracle
```

```
# Ganti IP di bawah dengan IP yang muncul di teks sambutan MobaXterm tadi
```

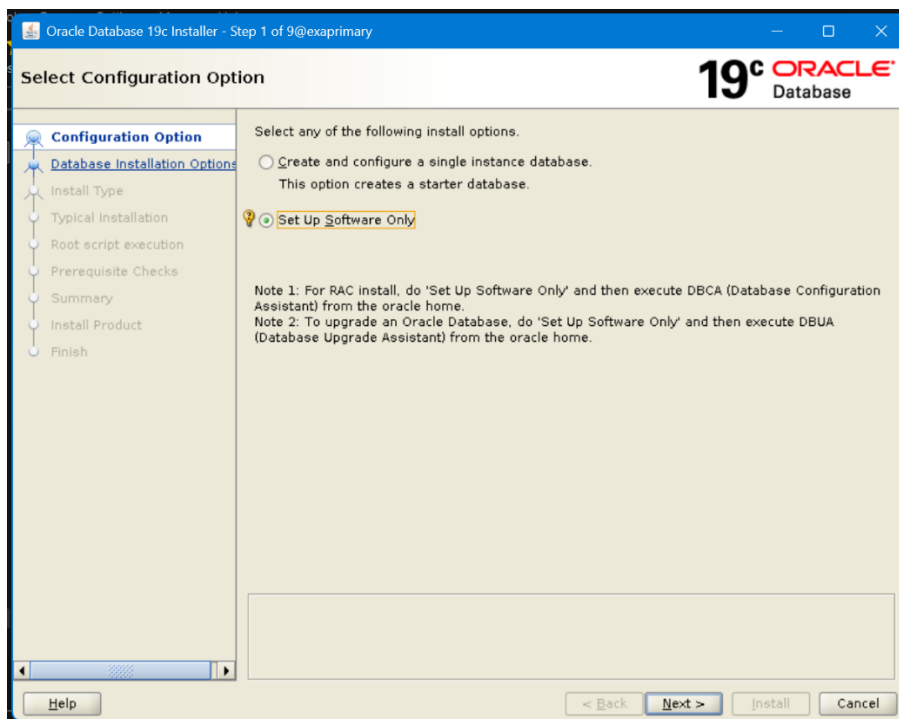
```
- export DISPLAY=192.168.56.1:0.0
```

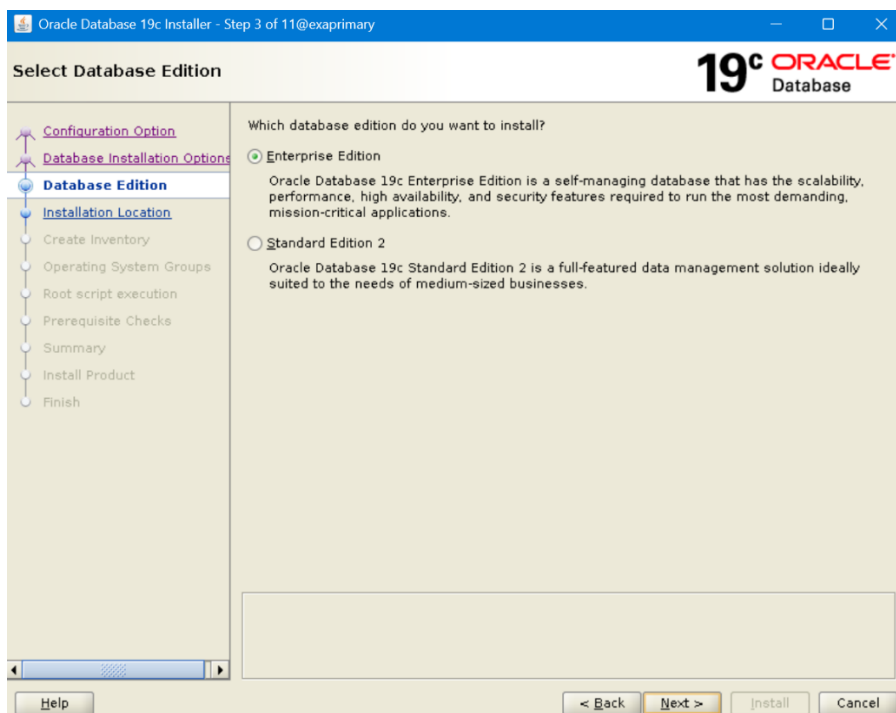
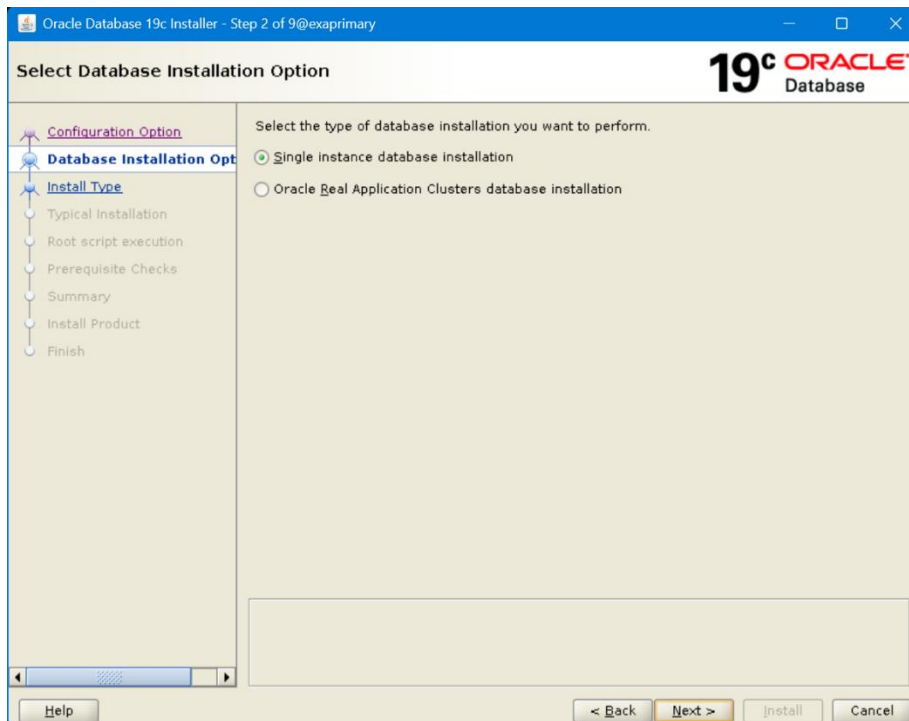
```
- xclock
```

41. Lalu run

```
cd /u01/app/oracle/product/19.0.0/dbhome_1
```

```
./runInstaller
```





Oracle Database 19c Installer - Step 4 of 11@exaprimary

Specify Installation Location

19c ORACLE Database

[Configuration Option](#)
[Database Installation Options](#)
[Database Edition](#)
[Installation Location](#)
[Create Inventory](#)
[Operating System Groups](#)
[Root script execution](#)
[Prerequisite Checks](#)
[Summary](#)
[Install Product](#)
[Finish](#)

Specify a path to place all Oracle software and configuration-related files installed by this installation owner. This location is the Oracle base directory for the installation owner.

Oracle base:

This software directory is the Oracle Database home directory.

Software location: /u01/app/oracle/product/19.0.0/dbhome_1

Oracle Database 19c Installer - Step 5 of 11@exaprimary

Create Inventory

19c ORACLE Database

[Configuration Option](#)
[Database Installation Options](#)
[Database Edition](#)
[Installation Location](#)
[Create Inventory](#)
[Operating System Groups](#)
[Root script execution](#)
[Prerequisite Checks](#)
[Summary](#)
[Install Product](#)
[Finish](#)

You are starting your first installation on this host. Specify a directory for installation metadata files (for example, install log files). This directory is called the "inventory directory". The installer automatically sets up subdirectories for each product to contain inventory data. The subdirectory for each product typically requires 150 kilobytes of disk space.

Inventory Directory:

Specify an operating system group whose members have write permission to the inventory directory (orainventory).

orainventory Group Name:

Oracle Database 19c Installer - Step 6 of 11@exaprimary

Privileged Operating System groups

19c ORACLE Database

Configuration Option
Database Installation Options
Database Edition
Installation Location
Create Inventory
Operating System Groups
Root script execution
Prerequisite Checks
Summary
Install Product
Finish

SYS privileges are required to create a database using operating system (OS) authentication. Membership in OS Groups grants the corresponding SYS privilege, eg. membership in OSDBA grants the SYSDBA privilege.

Database Administrator (OSDBA) group: oracle

Database Operator (OSOPER) group (Optional): oracle

Database Backup and Recovery (OSBACKUPDBA) group: oracle

Data Guard administrative (OSDGDBA) group: oracle

Encryption Key Management administrative (OSKMDBA) group: oracle

Real Application Cluster administrative (OSRACDBA) group: oracle

Help < Back Next > Install Cancel

Oracle Database 19c Installer - Step 7 of 11@exaprimary

Root script execution configuration

19c ORACLE Database

Configuration Option
Database Installation Options
Database Edition
Installation Location
Create Inventory
Operating System Groups
Root script execution
Prerequisite Checks
Summary
Install Product
Finish

During the software configuration, certain operations have to be performed as "root" user. You can choose to have the installer perform these operations automatically by specifying inputs for one of the options below. The input specified will also be used by the installer to perform additional prerequisite checks.

☐ Automatically run configuration scripts

☒ Use "root" user credential

Password:

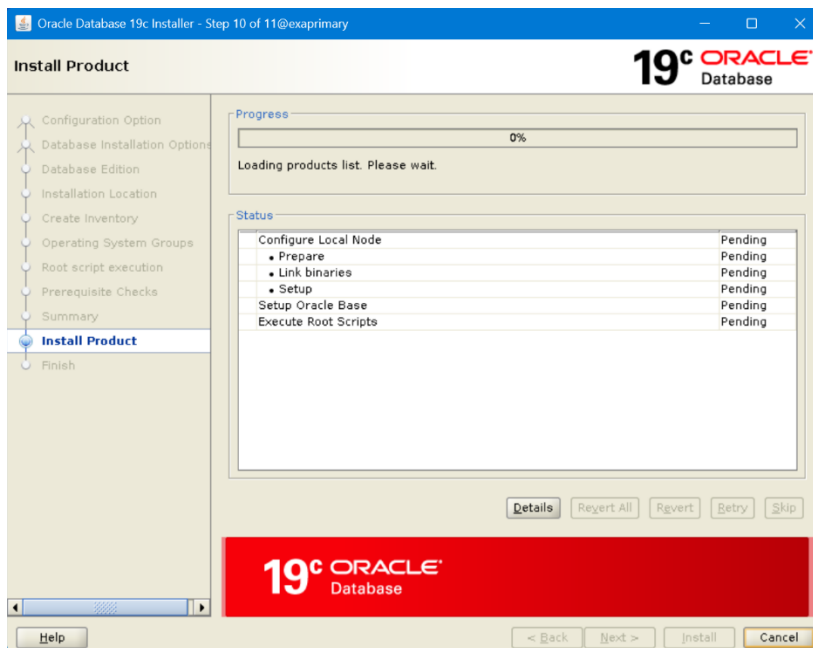
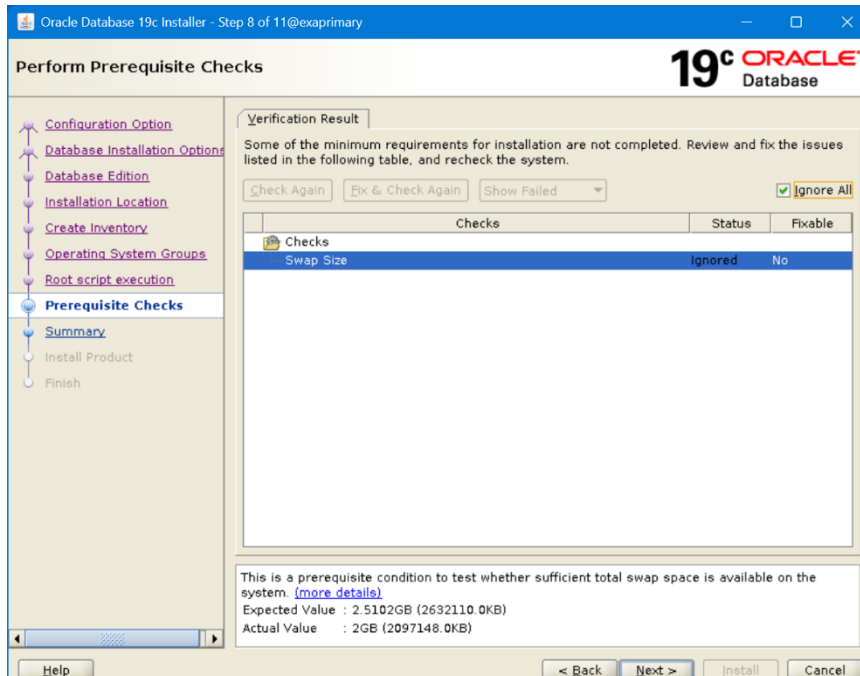
☐ Use sudo

Program path: /usr/bin/sudo Browse...

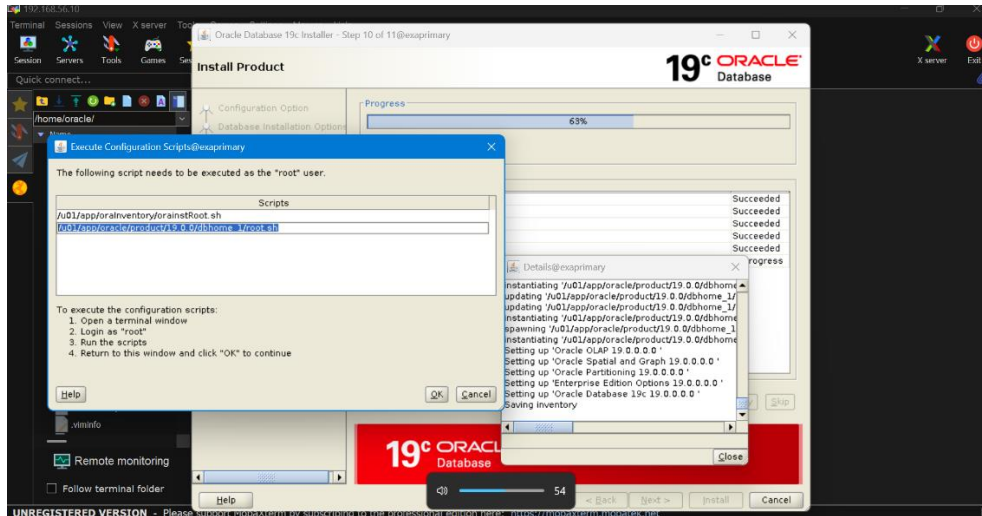
User name: oracle

Passwgrd:

Help < Back Next > Install Cancel



Duplicate tab, masuk ke user root lalu jalankan script tersebut



```
Last login: Wed May 27 21:32:42 2026
[oracle@exaprimary ~]$ su
Password:
[root@exaprimary oracle]# /u01/app/oraInventory/oraInstRoot.sh
Changing permissions of /u01/app/oraInventory.
Adding read,write permissions for group.
Removing read,write,execute permissions for world.

Changing groupname of /u01/app/oraInventory to oracle.
The execution of the script is complete.
[root@exaprimary oracle]# /u01/app/oracle/product/19.0.0/dbhome_1/root.sh
Performing root user operation.

The following environment variables are set as:
ORACLE_OWNER= oracle
ORACLE_HOME= /u01/app/oracle/product/19.0.0/dbhome_1

Enter the full pathname of the local bin directory: [/usr/local/bin]: orcl
Creating orcl directory...
Copying dbhome to orcl ...
Copying oraenv to orcl ...
Copying coraenv to orcl ...

Creating /etc/oratab file...
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Oracle Trace File Analyzer (TFA - Standalone Mode) is available at :
/u01/app/oracle/product/19.0.0/dbhome_1/bin/tfactl
Support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net
```

42. Lalu sebelum masuk ke dbca, atur environment variable terlebih dahulu

su - oracle

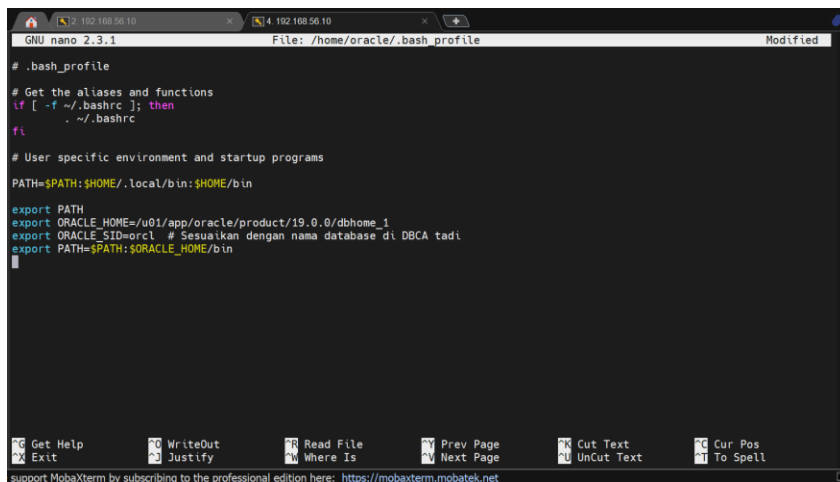
- nano ~/.bash_profile

Tambahkan baris ini di paling bawah:

export ORACLE_HOME=/u01/app/oracle/product/19.0.0/dbhome_1

export ORACLE_SID=orcl # Sesuaikan dengan nama database di DBCA tadi

export PATH=\$PATH:\$ORACLE_HOME/bin



```
GNU nano 2.3.1 File: /home/oracle/.bash_profile Modified
# .bash_profile

# Get the aliases and functions
if [ -f ~/.bashrc ]; then
    . ~/.bashrc
fi

# User specific environment and startup programs

PATH=$PATH:$HOME/.local/bin:$HOME/bin

export PATH
export ORACLE_HOME=/u01/app/oracle/product/19.0.0/dbhome_1
export ORACLE_SID=orcl # Sesuaikan dengan nama database di DBCA tadi
export PATH=$PATH:$ORACLE_HOME/bin
```

Simpan Ctrl+O, Enter dan Keluar (Ctrl+X).

Lalu aktifkan dengan:

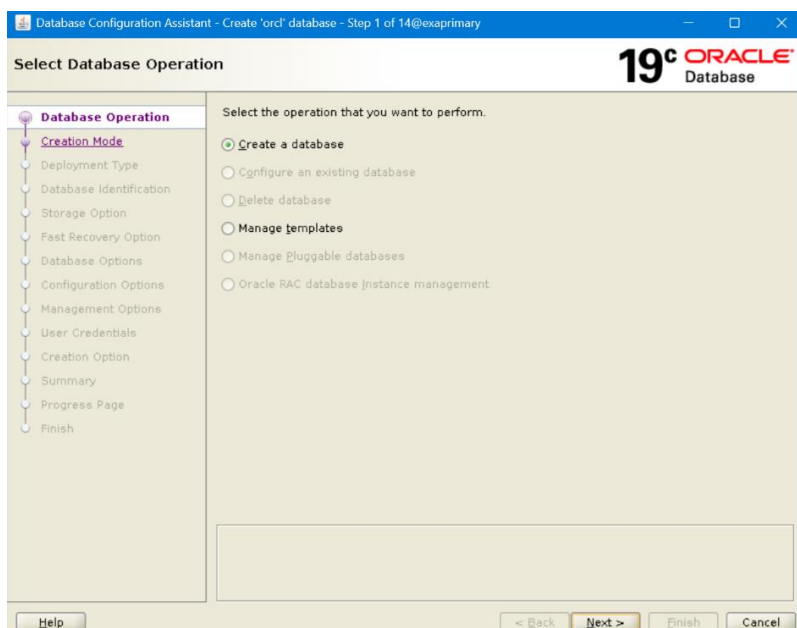
- source ~/.bash_profile

43. masuk ke oracle home

cd \$ORACLE_HOME

44. membuat database

dbca



Database Configuration Assistant - Create a database - Step 2 of 14@exaprimary

Select Database Creation Mode

19c ORACLE Database

Database Operation

- Creation Mode
- Deployment Type
- Database Identification
- Storage Option
- Fast Recovery Option
- Database Options
- Configuration Options
- Management Options
- User Credentials
- Creation Option
- Summary
- Progress Page
- Finish

☒ Typical configuration

Global database name:

Storage type:

Database files location:

Fast Recovery Area (FRA):

Database character set:

Administrative password:

Confirm password:

☐ Create as Container database

Pluggable database name:

☐ Advanced configuration

Messages:

Administrative password:[DBT-06208] The 'ADMIN' password entered does not conform to the Oracle recommended standards.

Database Configuration Assistant - Create 'orcl' database - Step 4 of 5@exaprimary

Progress Page

19c ORACLE Database

Database Operation

- Creation Mode
- Summary
- Progress Page
- Finish

Progress

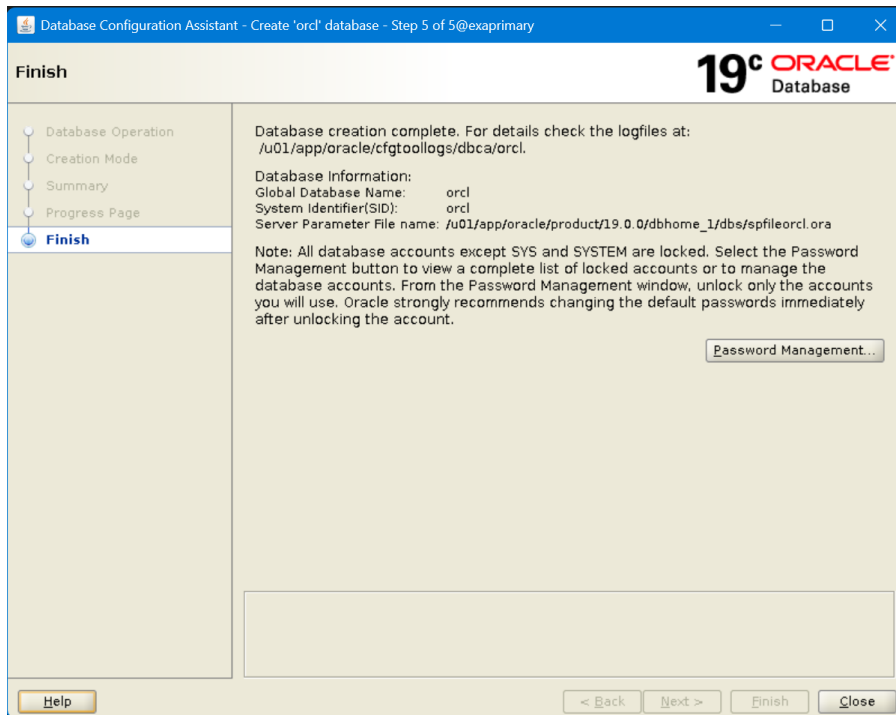
10%

Copying database files : In Progress

Status

Task	Status
DB Creation	In Progress
• Prepare for db operation	Succeeded
• Copying database files	In Progress
• Creating and starting Oracle instance	Pending
• Completing Database Creation	Pending
• Executing Post Configuration	Pending

DBCA Log Location: /u01/app/oracle/cfgtoollogs/dbca



Jika sudah seperti ini langsung close saja.

Dibuat oleh Naufal Ali Hilmi | Information Systems Student, Gunadarma University